OVER A LOCALLY NOETHERIAN ALGEBRAIC STACK, LOCALLY OF FINITE TYPE IMPLIES LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The presentation of a morphism of algebraic stacks by a square of schemes with smooth surjective vertical arrows. Used at the start of the counted proof (source lines 5668-5671). Reduces the claim about morphisms of stacks to a claim about the presenting morphisms of schemes.
- The statement for schemes that a morphism locally of finite type with locally Noetherian target is locally of finite presentation. Invoked inside the counted proof (source lines 5672-5675) as a cross-chapter pointer. Provides local finite presentation of the presenting morphism h.
- The definition of local finite presentation for a morphism of algebraic stacks. Used at the end of the countable part of the proof (source lines 5676-5678). Transfers the property from the presentation back to the morphism of stacks.

Source reference. Original lemma lemma-local-source-target: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3377–3377. Navigation pointer only: the counted text refers to the Stacks item lemma-local-source-target, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-noetherian-finite-type-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3847–3847. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-noetherian-finite-type-finite-presentation,

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which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks.*

(1) *If $\mathcal{Y}$ is locally Noetherian and f locally of finite type then f is locally of finite presentation.*
(2) *If $\mathcal{Y}$ is locally Noetherian and f of finite type and quasi-separated then f is of finite presentation.*

Proof. Assume $f : \mathcal{X} \to \mathcal{Y}$ locally of finite type and $\mathcal{Y}$ locally Noetherian. This means there exists a diagram as in Lemma lemma-local-source-target with h locally of finite type and surjective vertical arrow a. By Morphisms of Spaces, Lemma spaces-morphisms-lemma-noetherian-finite-type-finite-presentation h is locally of finite presentation. Hence $\mathcal{X} \to \mathcal{Y}$ is locally of finite presentation by definition. This proves (1). If f is of finite type and quasi-separated then it is also quasi-compact and quasi-separated and (2) follows immediately. □